

MLSecOps in Action

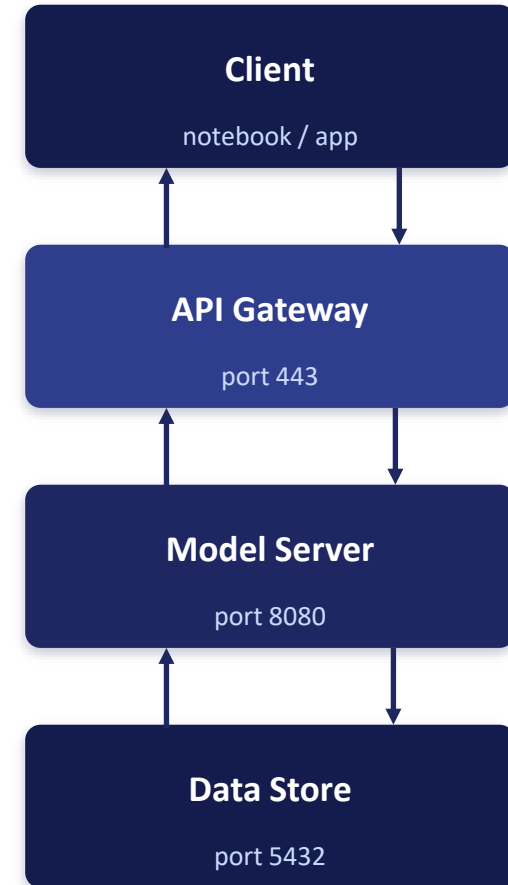
Networking Foundations

Core concepts to know before we open up iptables

Mohammad Fozouni

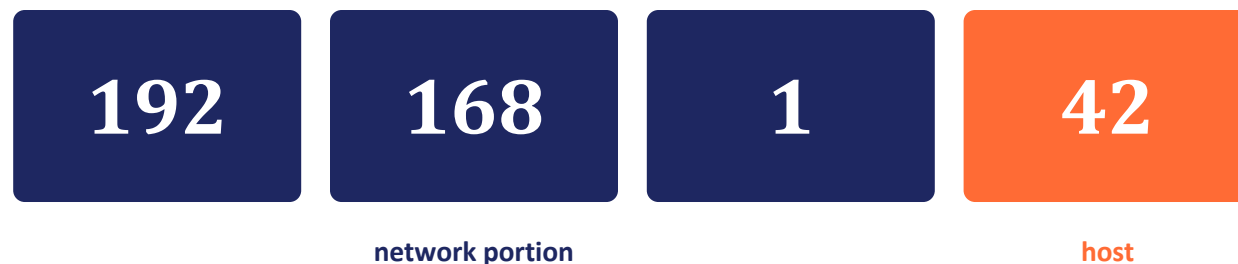
Why Networking Matters Here

- ML systems are network services. Training jobs, feature stores, and model endpoints all talk over the network.
- A model serving endpoint is, from a network's point of view, just another open port.
- Attackers look for the same things here as anywhere else: reachable services, weak boundaries, and unfiltered traffic.
- Everything we cover today is the vocabulary for which iptables rules are written in.



IP Addresses & Subnets

Every host needs an address and a way to know which hosts are "nearby"



An **IP address** is like a **postal address** for your computer on a network.

/24 subnet mask (255.255.255.0) — the first 24 bits identify the network

Private address ranges

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Not routable on the public internet — this is typically where your training clusters and internal model endpoints live.

IP Addresses & Subnets



network portion

host

Private address ranges

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Not routable on the public internet — this is typically where your training clusters and internal model endpoints live.

XXXXXXXX

XXXXXXXX

XXXXXXXX

XXXXXXXX

11000000

10101000

00000001

00101010

192 168 1 42

network portion

host

XXXXXXXX	XXXXXXXX	XXXXXXXX	XXXXXXXX
11000000	10101000	00000001	00101010

192

168

1

42

network portion

host

XXXXXXXX

XXXXXXXX

XXXXXXXX

XXXXXXXX

11000000

10101000

00000001

00101010

$$10101000 = 0 \times 2^0 + 0 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 0 \times 2^4 + 1 \times 2^5 + 0 \times 2^6 + 1 \times 2^7 = 8 + 32 + 128 = 168$$

192

168

1

42

network portion

host

XXXXXXXX

XXXXXXXX

XXXXXXXX

XXXXXXXX

11000000

10101000

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$$10101000 = 0 \times 2^0 + 0 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 0 \times 2^4 + 1 \times 2^5 + 0 \times 2^6 + 1 \times 2^7 = 8 + 32 + 128 = 168$$

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192

168

1

42

network portion

host

/24 subnet mask (255.255.255.0). The first 24 bits identify the network

*Full Range (including **network** and **broadcast**):*

192.168.1.0 ==> till ==> 192.168.1.255

*192.168.1.1 ==> till ==> 192.168.1.254 (**Usable**)*

So, 254 device can connect to this network

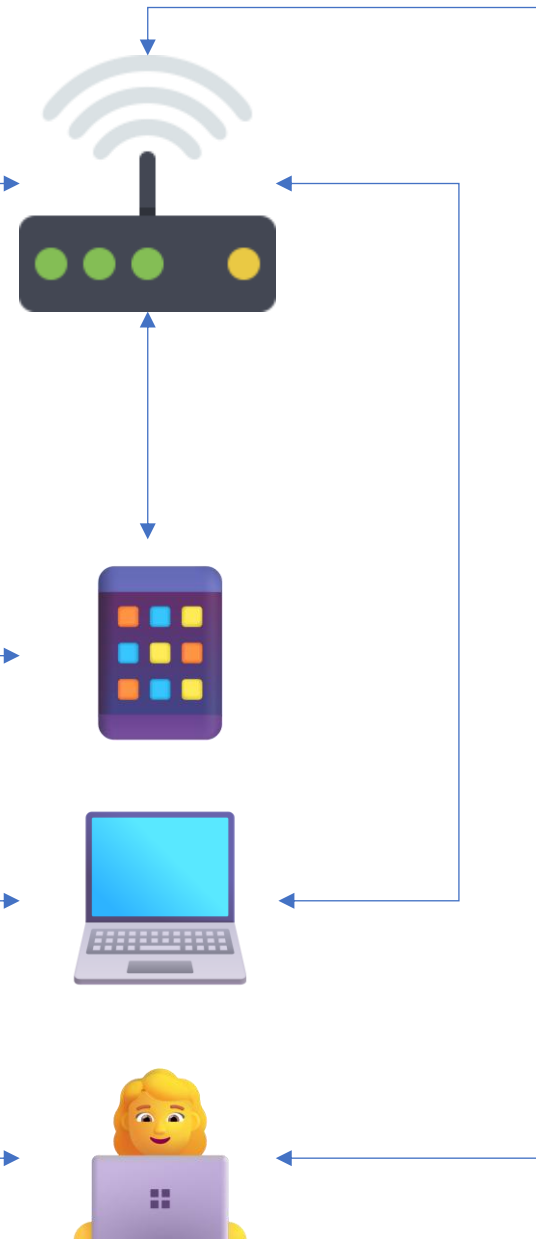
192	168	1	1
-----	-----	---	---

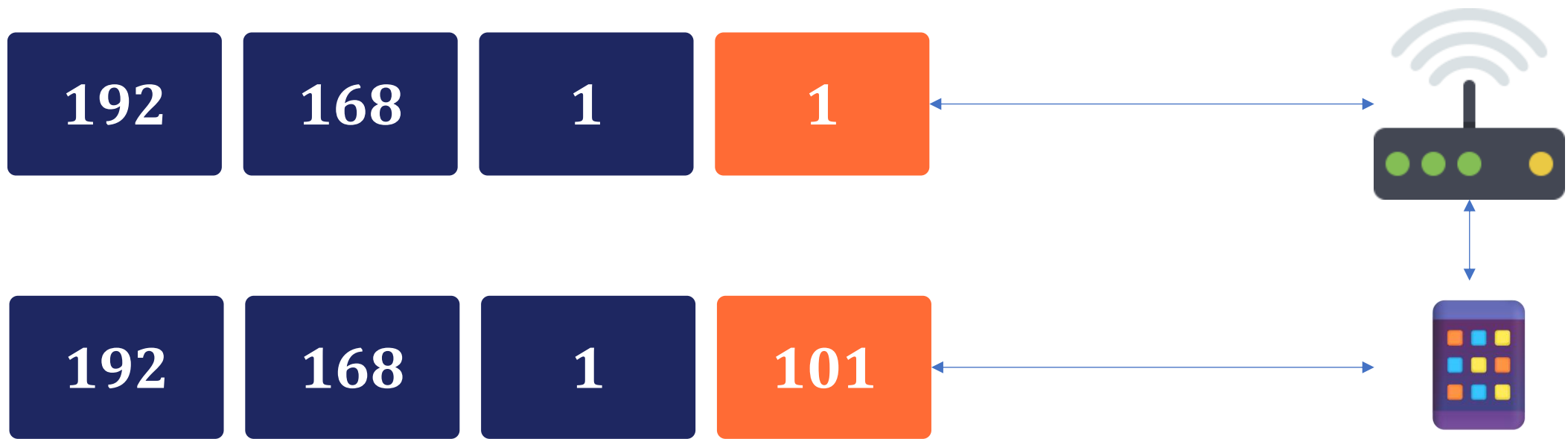
Home Network

192	168	1	101
-----	-----	---	-----

192	168	1	104
-----	-----	---	-----

192	168	1	???
-----	-----	---	-----





This is Public IP?

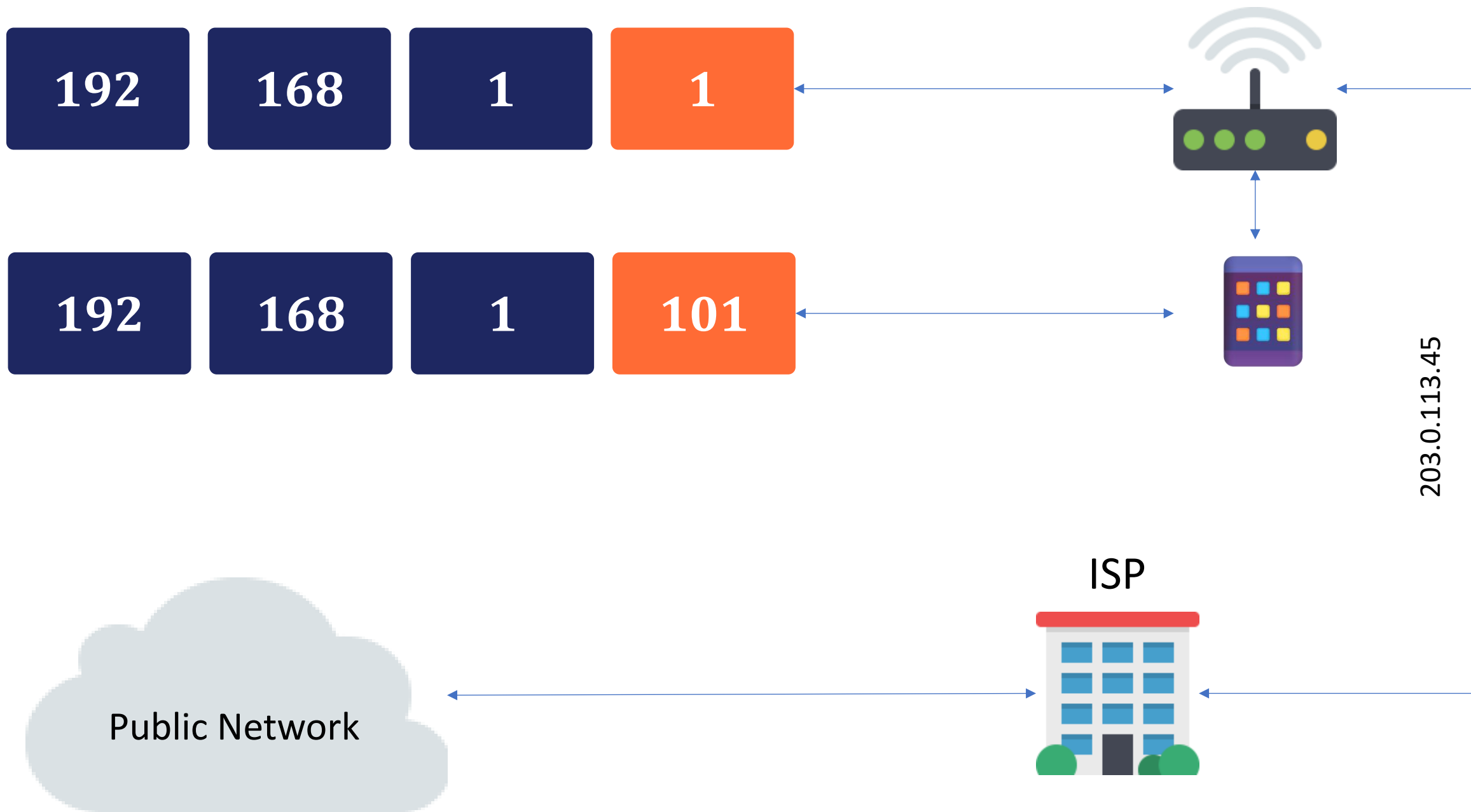
NO

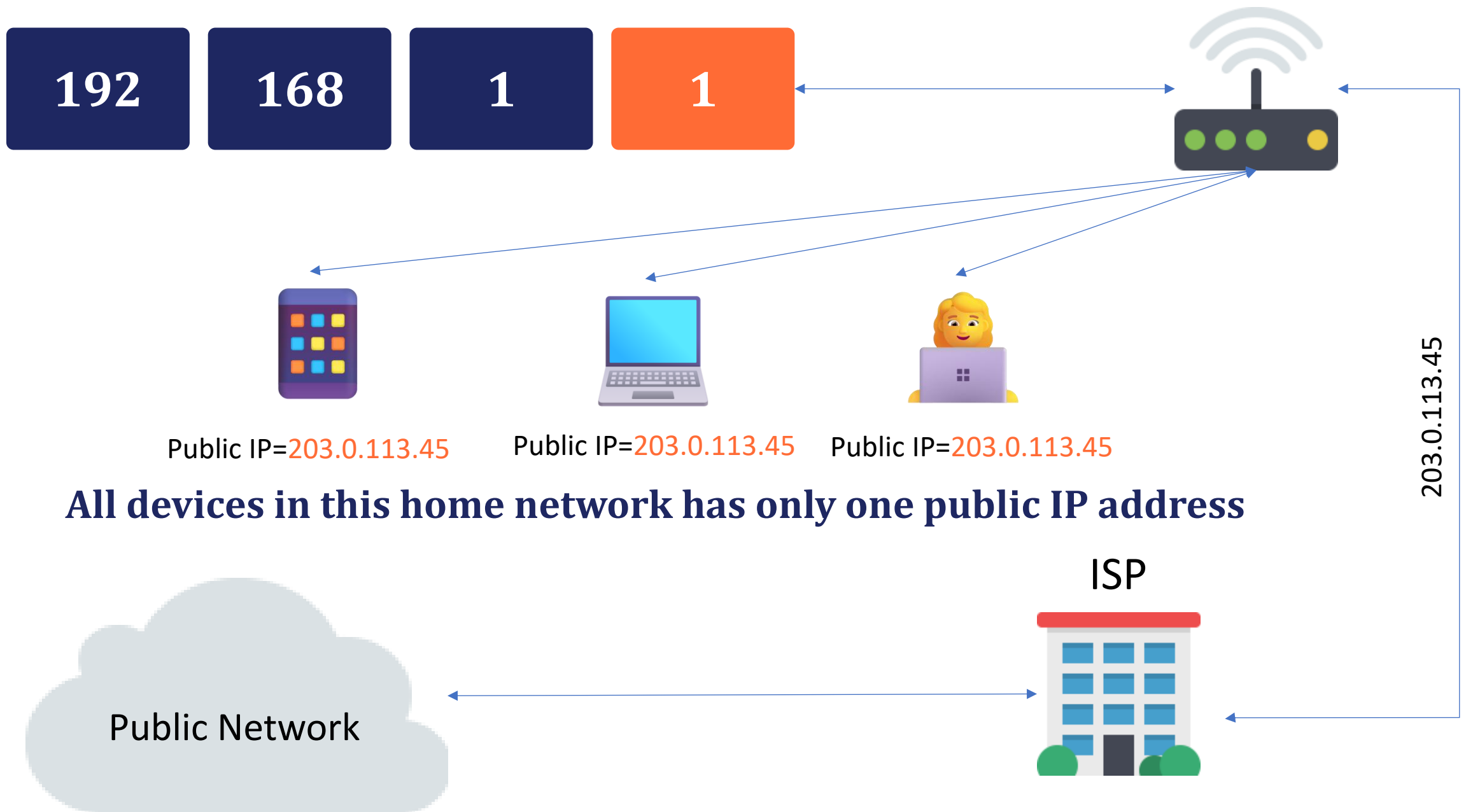
**So, how our devices get
to network publicly?**

Private address ranges

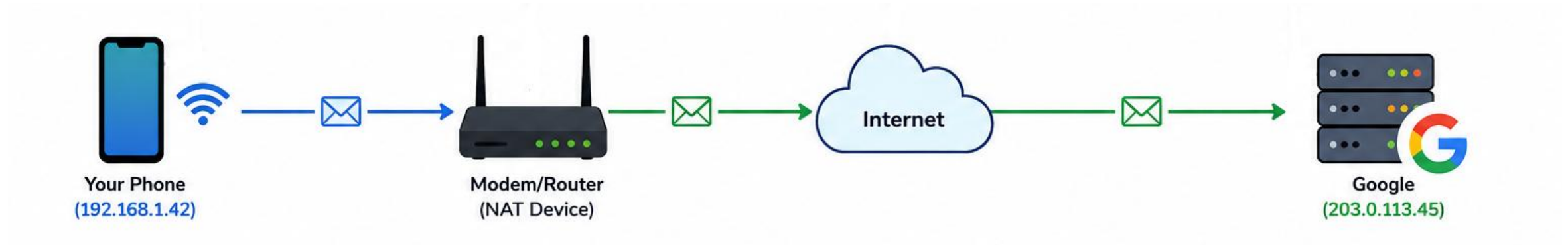
- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Not routable on the public internet — this is typically where your training clusters and internal model endpoints live.



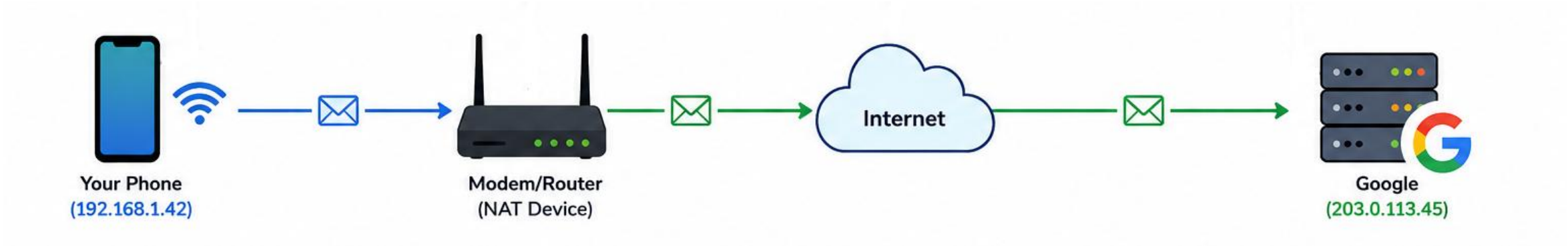


NAT or Network Address Translation



src ip: port	dst ip: port
192.168.1.42:54321	203.0.113.45:54321

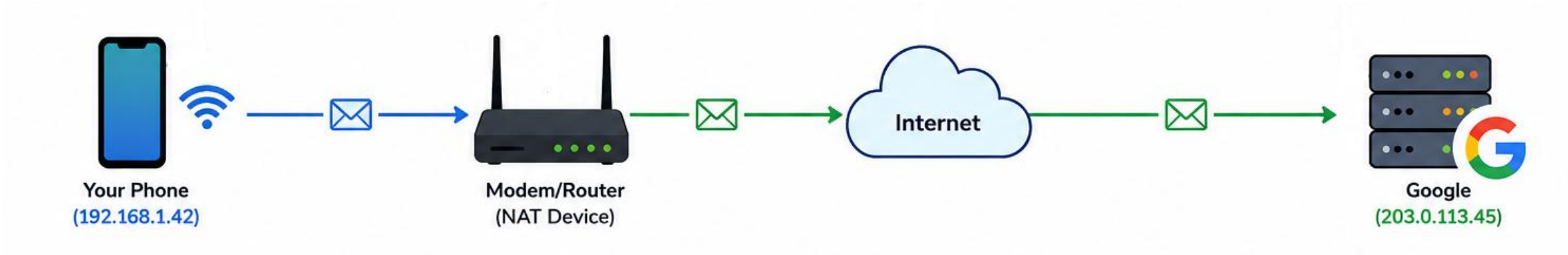
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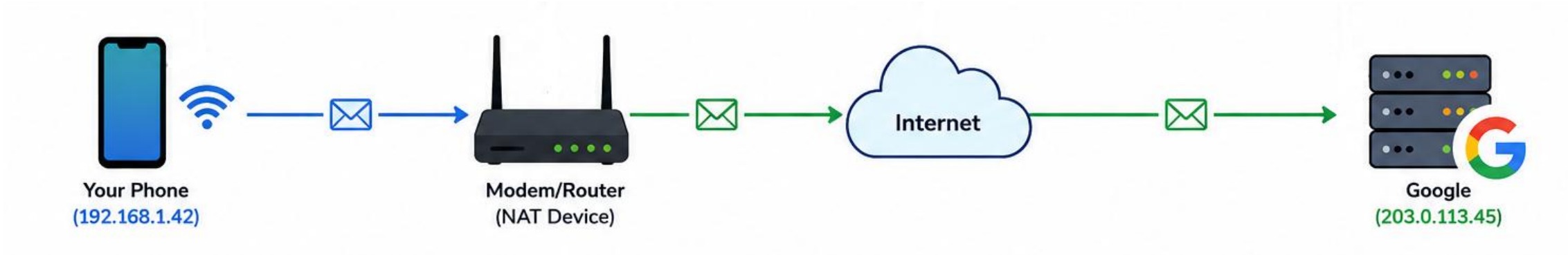
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src ip: port	dst ip: port
5.112.11.22:54321	203.0.113.45:54321

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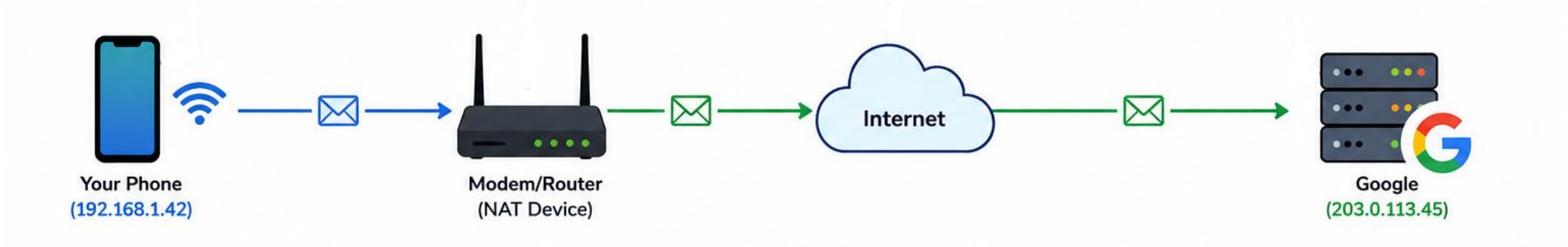
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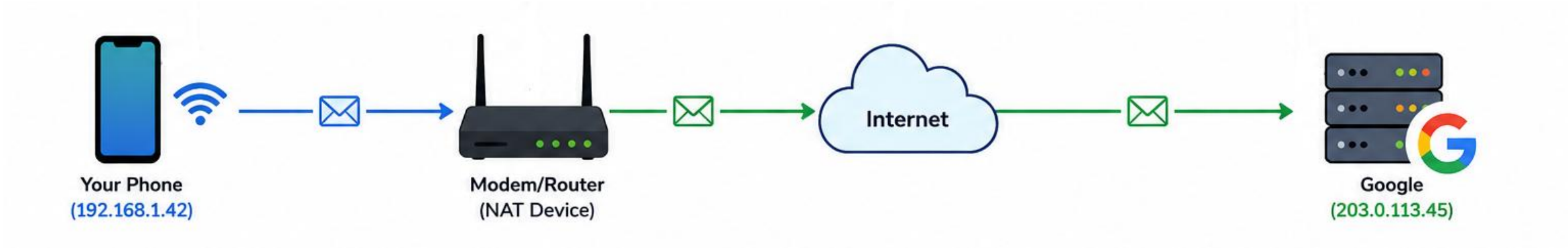
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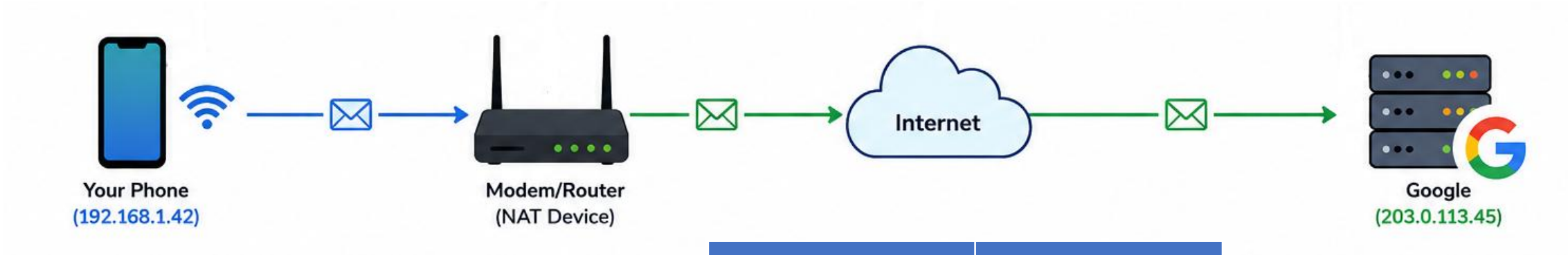
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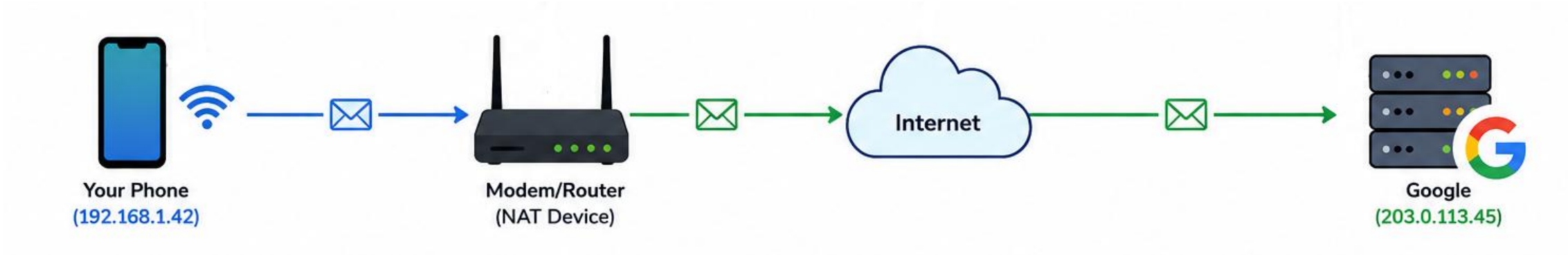
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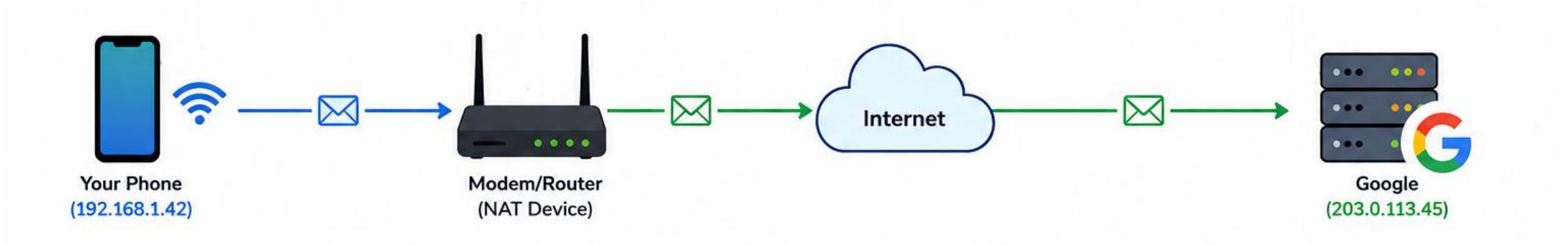
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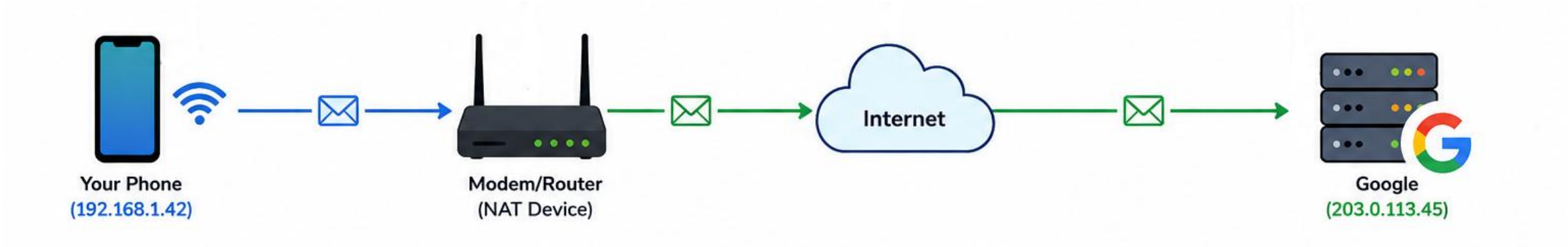
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NAT or Network Address Translation



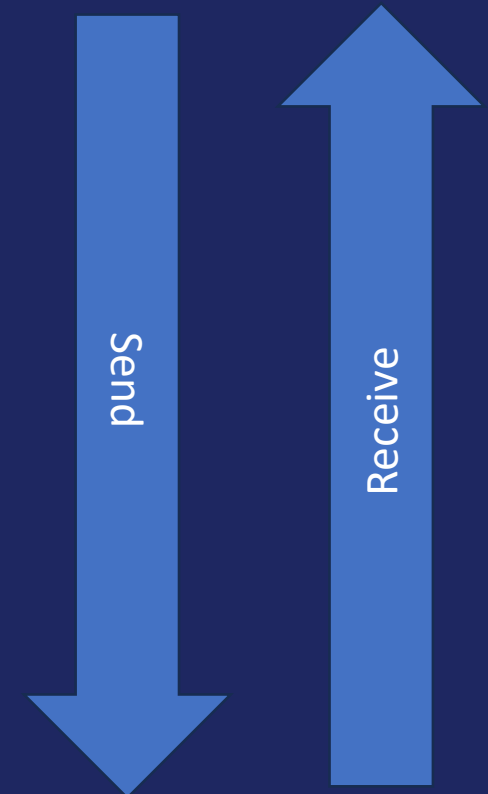
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203.0.113.45:54321	192.168.1.42:54321

src ip: port	dst ip: port
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The TCP/IP Model

Four layers describe how an application's data becomes signals on a wire

Application	HTTP, DNS, gRPC — the model's API calls live here
Transport	TCP / UDP — reliability and the port number
Internet	IP — logical addressing and routing between networks
Network Access	Ethernet, Wi-Fi — placing bits on the physical link



Each layer only cares about the header it owns.

Ports & Protocols

A port tells the OS which service on a host should receive the traffic

TCP

connection-oriented

- Handshake before data flows
- Guarantees delivery & order
- Used for: HTTP(S), SSH, model APIs

UDP

connectionless

- No handshake — just send
- Faster, no delivery guarantee
- Used for: DNS lookups, streaming, metrics

Common ports

22

SSH

80

HTTP

443

HTTPS

5432

PostgreSQL

8080 / 5000

Model-serving APIs

Anatomy of a Packet

Firewalls make decisions by reading headers, not the payload

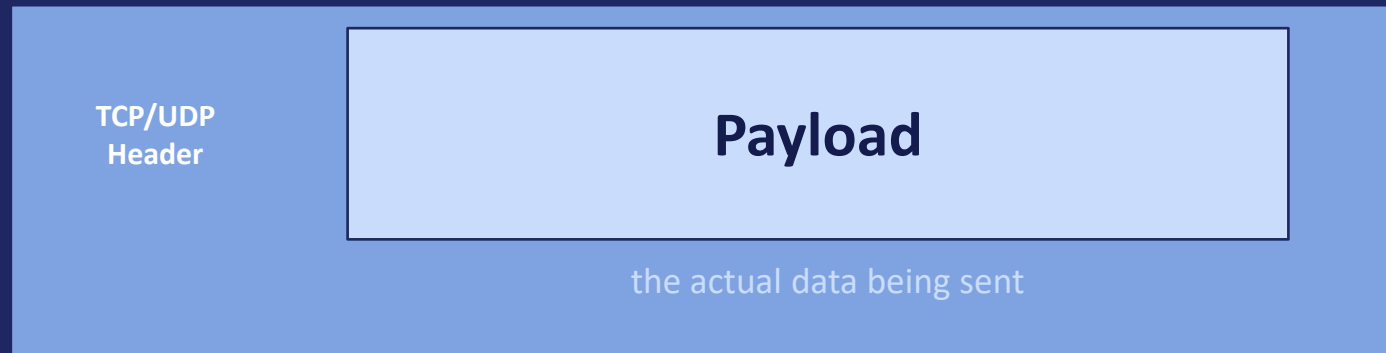
محتوای اصلی داخل پیام ما

Payload

the actual data being sent

Anatomy of a Packet

Firewalls make decisions by reading headers, not the payload



src / dst port

src port کدام اپ این را ارسال می کند؟

dst port کدام اپ این را دریافت خواهد کرد؟

Anatomy of a Packet

Firewalls make decisions by reading headers, not the payload



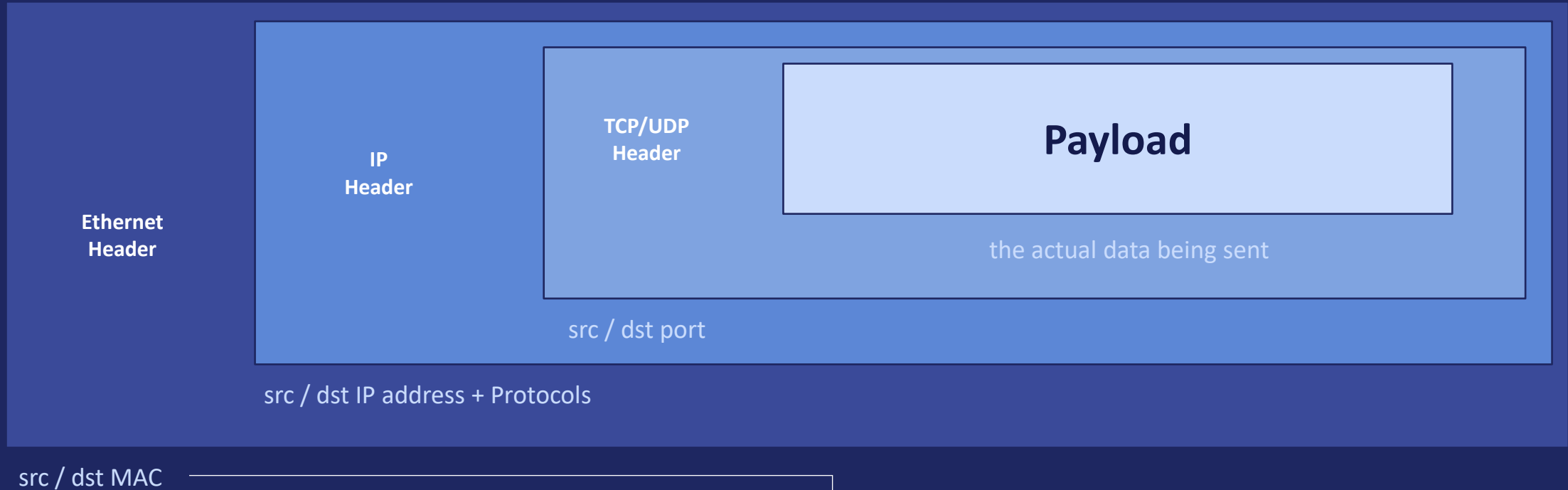
src / dst IP address + Protocols

src ip یعنی آدرس فرستنده نامه چیست

dst ip یعنی آدرس گیرنده نامه کجاست

Anatomy of a Packet

Firewalls make decisions by reading headers, not the payload

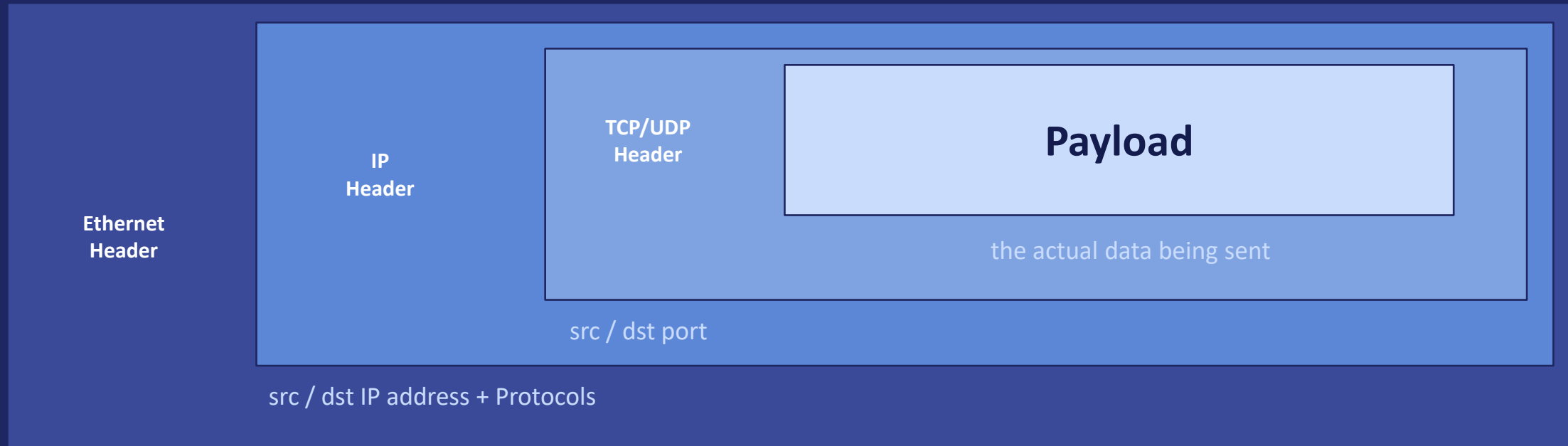


src MAC یعنی کدام دستگاه فیزیکی این نامه را جابجا می کند (تغییر می کند)

dst MAC آدرس دستگاه فیزیکی بعدی که این نامه را دریافت می کند (تغییر می کند)

Anatomy of a Packet

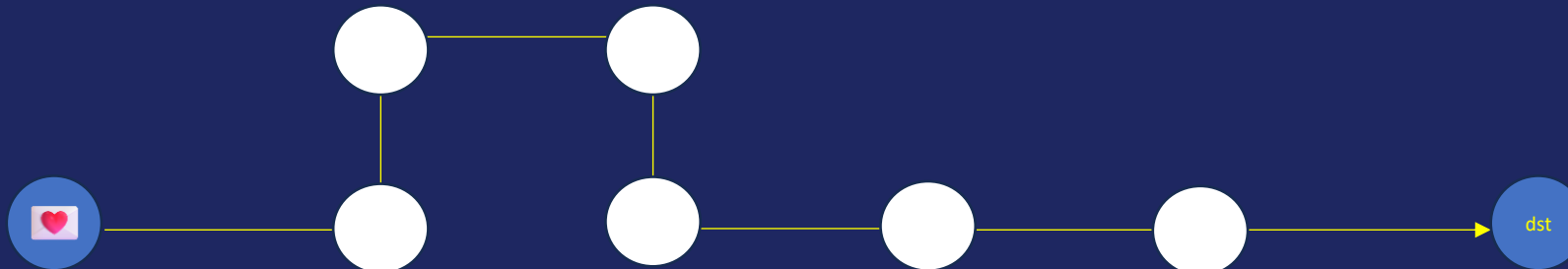
Firewalls make decisions by reading headers, not the payload



src / dst MAC

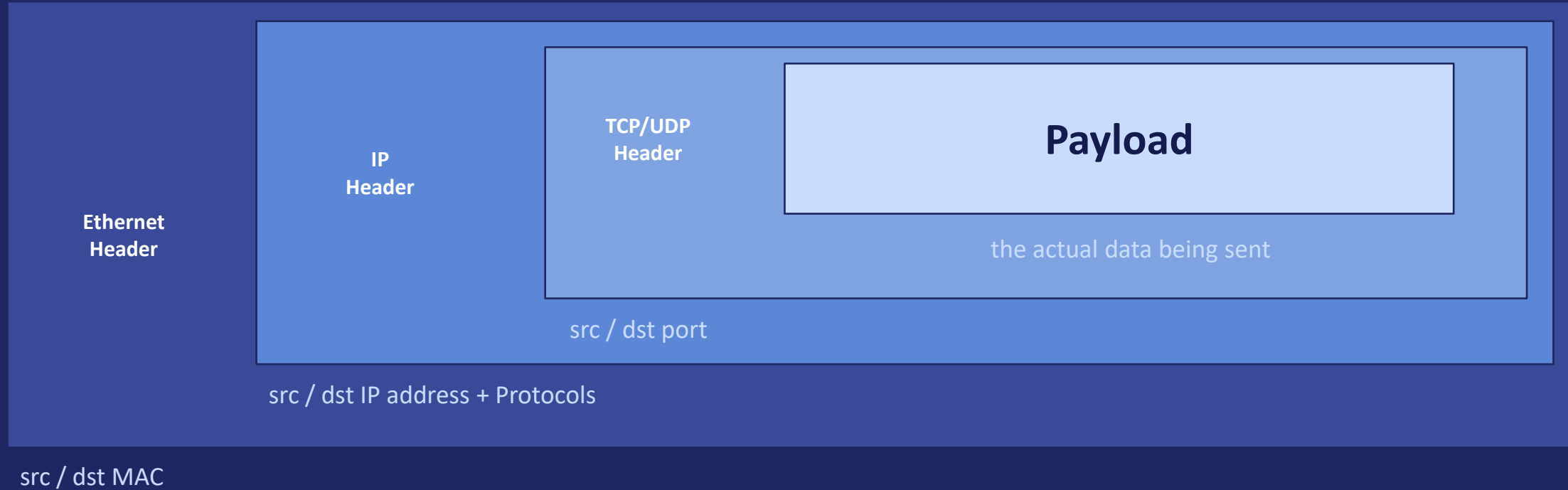
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Anatomy of a Packet

Firewalls make decisions by reading headers, not the payload



iptables reads the headers (**protocol, port, IP address**) to decide whether a packet is allowed through. It never needs to look inside the payload.

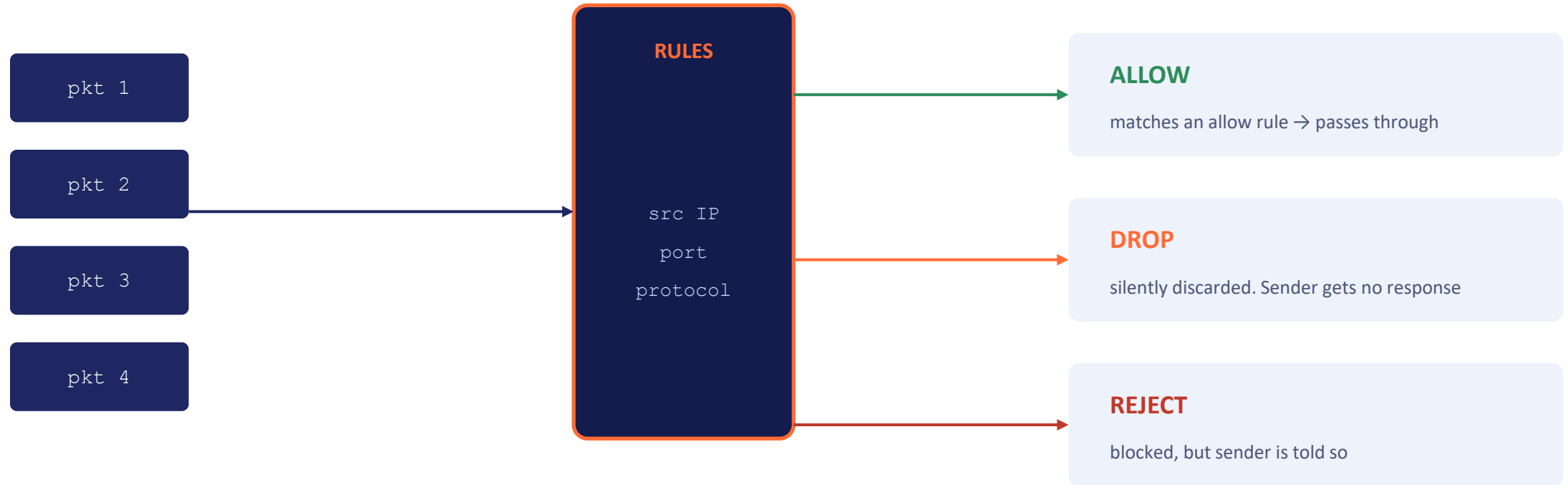
How Data Travels: A Request in Motion

A simplified look at a client calling a model-serving endpoint



What Is a Firewall?

A checkpoint that inspects packets and decides: allow, drop, or reject



***iptables** is one implementation of exactly this idea, running in the Linux kernel.*

Why This Matters for ML Systems

Network boundaries are part of the ML attack surface, not just an infra concern

1

Exposed inference endpoints

A model API left open to 0.0.0.0/0 can be scraped, abused, or used to extract the model itself.

2

Unrestricted egress

A compromised training job with open outbound access can exfiltrate data or weights quietly.

- **Ingress** = Traffic coming **in** to your network (e.g., users calling your API).
- **Egress** = Traffic going **out** from your network (e.g., your server calling the internet).

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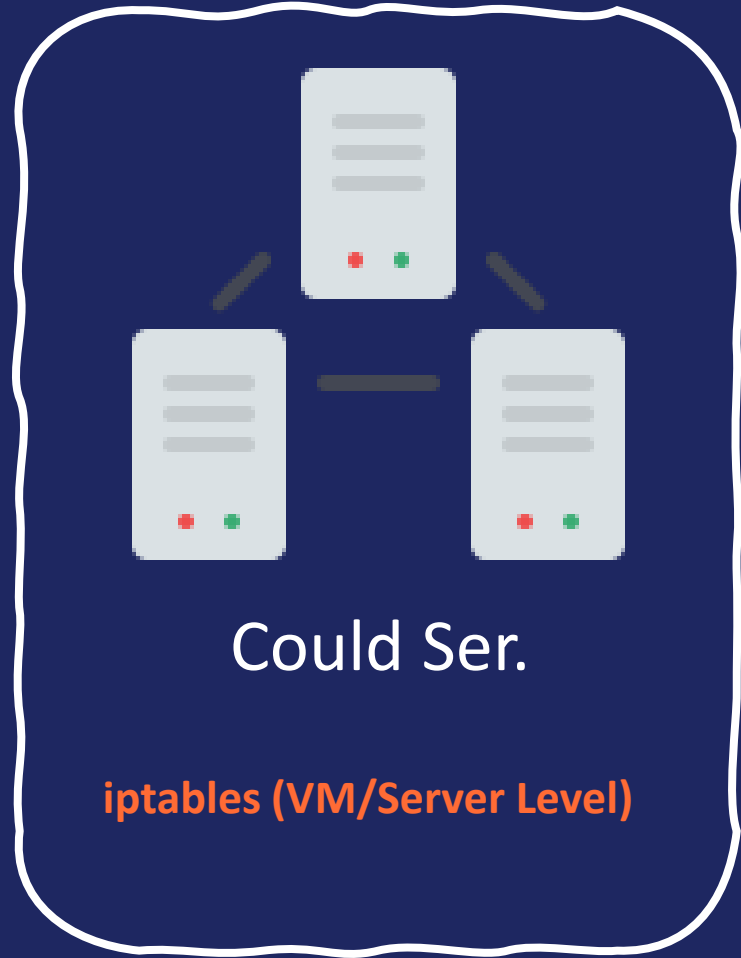
3

Lateral movement

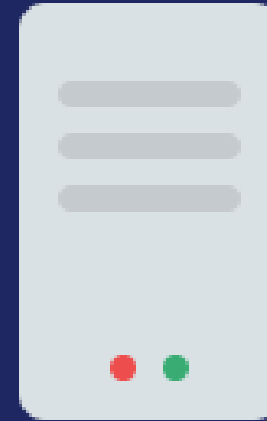
Flat networks let an attacker hop from a notebook to training data to production, unchecked.

A **flat network** is when **all devices can talk to all other devices** directly, with no segmentation or firewalls between them.

Always need iptables?



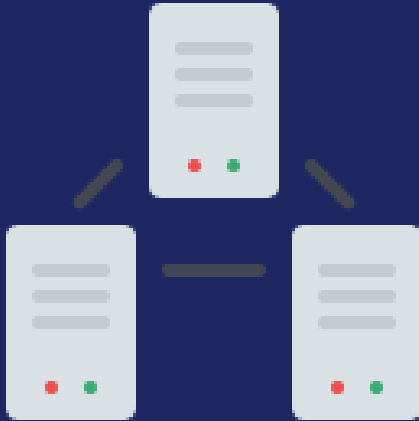
Cloud Firewall (Hypervisor Level)
Packets never reach the VM.



On Prem.

iptables (VM/Server Level)
Packets reach the VM but get
dropped inside.

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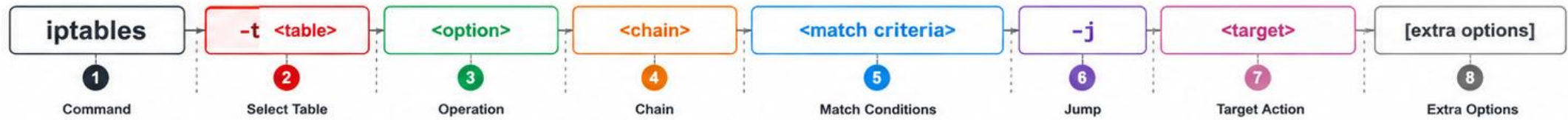
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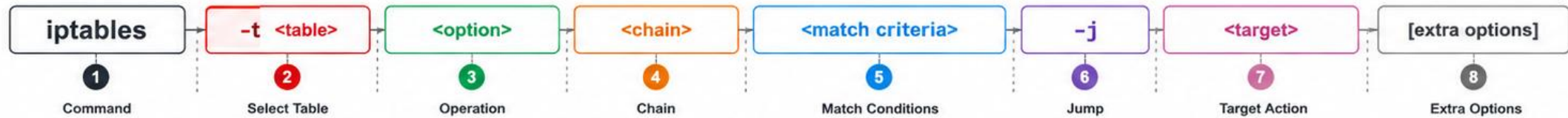
Packets reach the VM but get
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**Even on cloud panels, if we use iptables and cloud firewalls,
indeed we have two layers of defence. So always use iptables.**

iptables Command Structure



iptables Command Structure



2 Tables (-t)

Define which table to manipulate.

filter	Default table for packet filtering
nat	Network Address Translation
mangle	Packet alteration
raw	Raw table (early packet processing)

Default: **filter** (if not specified)

3 Options (Operation)

Define the operation to perform.

-A	Append rule(s) to the end
-D	Delete rule(s)
-I	Insert rule(s) at position
-R	Replace rule(s)
-L	List rules
-F	Flush (delete all rules)
-Z	Zero counters
-N	Create new chain
-X	Delete user-defined chain
-P	Set default policy
-E	Rename chain

Most common: **-A**, **-D**, **-L**

4 Chains

Define the chain in the table.

Built-in Chains (filter table)

INPUT	Packets destined for local system
OUTPUT	Packets generated by local system
FORWARD	Packets being routed through

Built-in Chains (nat table)

PREROUTING	Before routing decision
INPUT	After routing (local system)
OUTPUT	Before routing (local system)
POSTROUTING	After routing decision

Use **-L --line-numbers** to see chain rules

5 Match Criteria

Define conditions to match packets.

-s, --source <IP/CIDR>	Source IP address
-d, --destination <IP/CIDR>	Destination IP address
-p, --protocol <proto>	Protocol (tcp, udp, icmp, ...)
--sport <port>	Source port (for tcp/udp)
--dport <port>	Destination port (for tcp/udp)
-i, --in-interface <iface>	Incoming interface
-o, --out-interface <iface>	Outgoing interface
-m, --match <module>	Match extension module

...and many more (tcp, udp, icmp, state, etc.)

Multiple match criteria can be combined

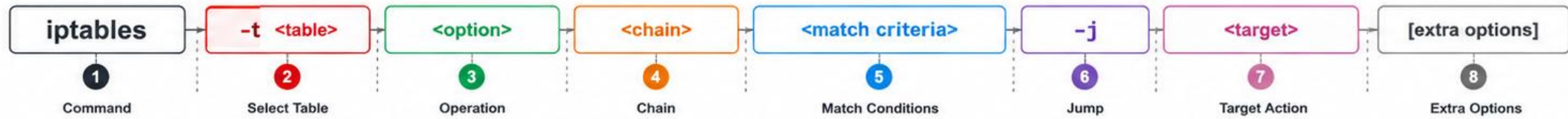
6 Jump (-j)

Specify the target for matched packets.

ACCEPT	Accept the packet
DROP	Silently drop the packet
REJECT	Drop and send error
RETURN	Return to calling chain
LOG	Log the packet
DNAT	Destination NAT
SNAT	Source NAT
MASQUERADE	IP masquerading (NAT)
REDIRECT	Redirect to local port

Target actions determine packet fate

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Target actions determine packet fate

7 Target Details



ACCEPT

Accept the packet and stop processing further rules.



DROP

Drop the packet silently. No response sent to sender.



REJECT

Drop the packet and send an error message (ICMP error).



LOG

Log packet details (used with **--log-prefix** or **--log-level**).



RETURN

Stop processing and return to the calling chain.



DNAT

Change destination IP address (used in nat table).



SNAT

Change source IP address (used in nat table).



MASQUERADE

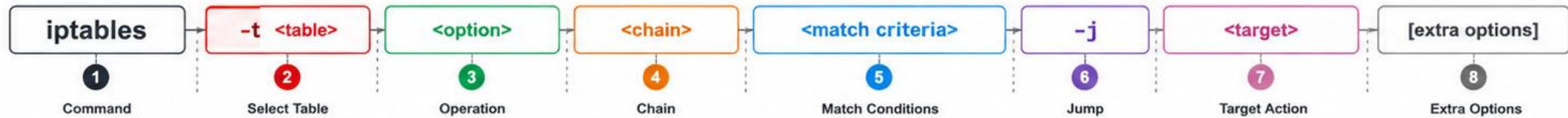
Dynamic source NAT (used for internet sharing).



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SNAT

Change source IP address (used in nat table).



MASQUERADE

Dynamic source NAT (used for internet sharing).



REDIRECT

Redirect packet to local system port.

8 Extra Options (Common)



--line-numbers (-L)

Show rule numbers when listing.



-v, --verbose

Verbose output (visible counters).

123

-n, --numeric

Show numeric addresses and ports.



--table <table>

Specify table (alternative to -t).



--modprobe=<command>

Use alternative command for module loading.



--wait

Wait for the xtables lock.



--wait-interval <sec>

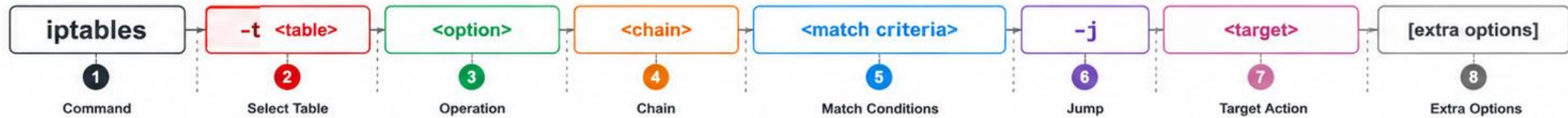
Wait time between retries.



--wait-timeout <sec>

Maximum wait time.

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Most common: **-A**, **-D**, **-L**

4 Chains

Define the chain in the table.

Built-in Chains (filter table)

INPUT	Packets destined for local system
OUTPUT	Packets generated by local system
FORWARD	Packets being routed through

Built-in Chains (nat table)

PREROUTING	Before routing decision
INPUT	After routing (local system)
OUTPUT	Before routing (local system)
POSTROUTING	After routing decision

Use **-L --line-numbers** to see chain rules

5 Match Criteria

Define conditions to match packets.

-s, --source <IP/CIDR>	Source IP address
-d, --destination <IP/CIDR>	Destination IP address
-p, --protocol <proto>	Protocol (tcp, udp, icmp, ...)
--sport <port>	Source port (for tcp/udp)
--dport <port>	Destination port (for tcp/udp)
-i, --in-interface <iface>	Incoming interface
-o, --out-interface <iface>	Outgoing interface
-m, --match <module>	Match extension module
...and many more (tcp, udp, icmp, state, etc.)	

Multiple match criteria can be combined

6 Jump (-j)

Specify the target for matched packets.

ACCEPT	Accept the packet
DROP	Silently drop the packet
REJECT	Drop and send error
RETURN	Return to calling chain
LOG	Log the packet
DNAT	Destination NAT
SNAT	Source NAT
MASQUERADE	IP masquerading (NAT)
REDIRECT	Redirect to local port

Target actions determine packet fate

7 Target Details



ACCEPT

Accept the packet and stop processing further rules.



DROP

Drop the packet silently. No response sent to sender.



REJECT

Drop the packet and send an error message (ICMP error).



LOG

Log packet details (used with **--log-prefix** or **--log-level**).



RETURN

Stop processing and return to the calling chain.



DNAT

Change destination IP address (used in nat table).



SNAT

Change source IP address (used in nat table).



MASQUERADE

Dynamic source NAT (used for internet sharing).



REDIRECT

Redirect packet to local system port.

8 Extra Options (Common)



--line-numbers (-L)

Show rule numbers when listing.



-v, --verbose

Verbose output (visible counters).

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-n, --numeric

Show numeric addresses and ports.



--table <table>

Specify table (alternative to **-t**).



--modprobe=<command>

Use alternative command for module loading.



--wait

Wait for the xtables lock.



--wait-interval <sec>

Wait time between retries.



--wait-timeout <sec>

Maximum wait time.



Example Command:

iptables -t filter -A INPUT -p tcp -s 192.168.1.0/24 --dport 22 -j ACCEPT

Meaning:

In the filter table, append a rule to the INPUT chain that accepts TCP packets from 192.168.1.0/24 destined for port 22 (SSH).

We now have the vocabulary:

addresses, ports, packets, private Ips, public Ips, NAT and rules.

Next: let's goo for iptables

Writing and reading real firewall rules on Linux